RAFA PRAMUDYA SUSANTO

AI & COMPUTER VISION / 3D PHYSICS SIMULATION

+62 821-1199-6769 · Bekasi, Jawa Barat, Indonesia · rafapramudya.dev

PROFILE

High-school developer focused on computer vision and real-time 3D simulation. I build systems that read the physical world: a gesture and object detection pipeline running on-device, and a rigid-body gravity sandbox with a hand-written contact solver.

Comfortable across the stack: OpenCV and landmark models on one side, WebGL and accessible React interfaces on the other. Looking for internships and open-source collaborations where the frame budget is real.

SELECTED PROJECTS

AI Gesture & Object Detection

Python · OpenCV · MediaPipe · TensorFlow Lite

A computer-vision pipeline that tracks hand landmarks and detects everyday objects from a live camera feed, then maps the result to gestures. Everything runs on-device at 31 FPS.

3D Physics Engine & Gravity Simulation

TypeScript · WebGL · Three.js · GLSL

A 3D rigid-body sandbox with a from-scratch integrator, impulse-based collision response, and a live parameter inspector. Built to stay stable at 60 FPS while the scene gets crowded.

TECHNICAL SKILLS

Languages	Python, TypeScript / JavaScript, C++, GLSL, SQL
Vision	OpenCV, MediaPipe, TensorFlow Lite, PyTorch, NumPy
Simulation	Three.js, WebGL, rigid-body dynamics, collision & constraints
Web	Next.js App Router, React, Tailwind CSS, Node.js, Git

EDUCATION

SMA Negeri 9 Bekasi

Mathematics & Computer Science · 26th Class

CERTIFICATES & HONOURS

- 1st Place — KSN Sub-district Mathematics 2020 — Bekasi City Education Office, March 2020
- 1st Place — PRF UI '23 Robotics — Universitas Indonesia, Oct 2025
- 1st Place — TFC '21 Quiz Bee — SMAN 1 Tambun Selatan, May 2026
- OSN Informatics 2026 Finalist — Puspresnas, June 2026 - Sep 2026

PLACEHOLDER RESUME — replace public/resume-en.pdf with your own CV. Generated by scripts/generate-resume.mjs.